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Scalable Aspects Learning for Intent-aware Diversified Search on Social Networks

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ABSTRACT Search result diversification on networks aims at selecting a set of representative nodes in response to a given query node so that the result is able to meet users' ambiguous query intents. Previous work mainly tackles this problem based on global diversity metrics, such as the expansion ratio and the expanded relevance, according to which the potential diversity needs of different query are interpreted as an unchanged criterion. While with various side information in real-world social networks, the intents of users often have more than one interpretation underlying the same query. In this paper, we therefore adopt an intent-aware perspective on this problem, based on network representation learning. With the hypothesis that a search result being aware of multiple intents of query is more likely to satisfy the information needs of users, we propose an intent-aware method that first encodes the possible query aspects and nodes as vectors, and then diversifies the search result based on these vectors. In particular, we present *aspect2vec*, a scalable and flexible network representation learning model, which maps nodes into low-dimensional vector spaces while preserving the network structure, the node attribute, and the query-oriented proximity. An attribute augmented sampling approach is proposed to sample corpus for the three contexts to train the model. Finally, we perform a comprehensive evaluation on our method with various baselines. The results show that our proposed method outperforms the state-of-the-art diversification algorithms.

INDEX TERMS Diversification, Graph search, Intent-aware, Network representation learning.

I. INTRODUCTION

Social networks, such as Facebook and Twitter, have changed the way people communicate and interact with each other and have turned into a tool for people to understand one's life status, personal interests and social relationships. With the increasing richness of social network resources, people are increasingly willing to query information from billions registered users on social networking websites [1], [2]. These queries, however, are often ambiguous and have more than one interpretation due to challenges from big data [3], [4], thus social networking service providers have to focus on understanding users' query intents in order to meet their ambiguous complex information needs. Search result diversification as a means to tackle ambiguity and eliminate redundancy plays a vital role in many research areas, such as textual retrieval [5], recommender systems [6] and graph search [7]. A good understanding of diversified search on

social networks is also of significance to further design applications such as link prediction [8], citation recommendation [9], and mining influential nodes [10].

For example, in Fig. 1, when searching for the name "Elon Musk" in Twitter¹, besides to the exact matching user "Elon Musk", Twitter also provides some other similar users as recommendations to let query users more easily find their interest targets to follow. To achieve sustainable development of their websites [11], a desirable result of such recommended list should be diverse so that users with different backgrounds can find their interest targets semantically [47], [48].

Existing approaches for diversifying search result on networks typically build on implicit representation of the query intent and are based upon some global diversified mechanism

¹www.twitter.com, an online social networking website.

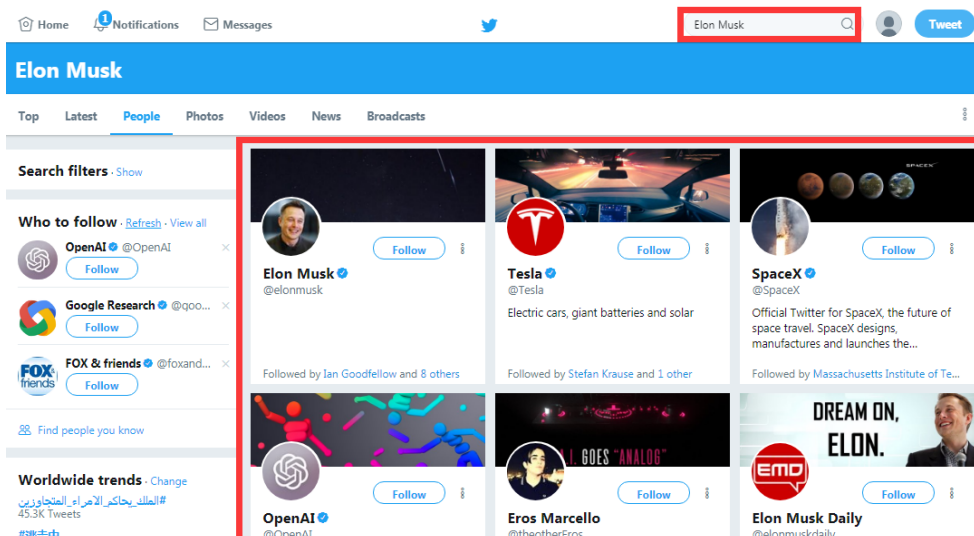


FIGURE 1: An example of user recommendation when searching "Elon Musk" in Twitter

or goodness, such as the rich-gets-richer mechanism [12], the direction-aware goodness [13] and the neighbour expansion ratio [7]. These methods try to identify the ‘correct’ interpretation behind any queries, without taking into account the intent of the query user itself. However, it is questionable whether users are actually interested in such nodes with a high global diversified goodness. In addition, existing diversified search algorithms focus on pure networks only and totally ignore the fact that nodes in real-world social networks are often associated with a rich set of features and attributes, which also provide rich complementary sources of information that should be used for uncovering, understanding and exploiting the latent diversified structure [49], [50].

In this paper we study the problem of diversifying search result on social networks. In particular, given a relation network with node attributes G , a query node q and result size k , our goal is searching for a result set of k nodes that are relevant to q and diverse to each other.

In this problem, a query is an expression of a user’s information need described by one node. With various side information of social networks, the possible intents underlying a query can be represented as multiple query aspects, which are the multiple subtopics or communities associated with such query. For instance, "Tesla" and "SpaceX" are two possible aspects of query "Elon Musk" in the example of Fig. 1. Existing diversification approaches in the field of text retrieval typically model query aspects according to the query log [14], the document topic [5] etc. In real social network scenarios, however, the actual aspects of the query are unknown, nor is the relevance of each retrieved node to each query aspect. Hence, it is challenging to effectively specify the query aspects in giving an arbitrary query user. In this paper, to address the problem of diversifying search result on social networks, we propose a novel intent-aware method which consists of two main steps. In the first step, the query

aspects as well as network nodes are represented as unified vectors. In the second step, two diversification algorithms are applied to select representative nodes according to these vectors.

Network Representation Learning (NRL), aiming to learn low-dimensional representations of nodes in networks, has attracted considerable attention in many real applications of networks. With the idea of node vectorization, we propose to utilize the NRL model to infer unknown latent aspects for the query and represent nodes. In the past decades, many NRL models have been proposed, most of which try to globally optimize a feature representation model that preserves the contextual information among all the participating objects [15]–[18]. However, these global optimized techniques will return a constant representation of nodes in any query, resulting in the fact that local important aspects of the query cannot be explicitly expressed, and conversely, marginal aspects may be overemphasized due to the heterogeneity of networks. This is the so-called ‘resolution problem’ in community detection research [19], [20]. This problem calls for a more flexible model that can map from retrieved nodes to feature spaces of query with different resolution and different number of aspects, so that the search algorithm can identify representative nodes precisely meeting the query intent. Further, to obtain diversified results for a new query, the NRL model need to relearn a new representation of nodes, which is costly for frequent searches in large scale social networks, and leads to new challenge for our network representation modelling.

To address the above challenges, we propose *aspect2vec*, a scalable and flexible NRL model that incorporates *query-oriented proximity* context to the optimization objective, enabling nodes to be encoded w.r.t. aspects of the query node according to their similarities to the query. More precisely, our *aspect2vec* model minimizes a loss function of preserving network structure, node attributes and query-oriented

proximity contexts to learn low-dimensional feature representation vectors of nodes, then the diversified result can be obtained according to these vectors through existing ripe algorithms of search result diversification. To train the model, we utilize the attribute augmented random walks approach to sample corpus for capturing the contextual information of network, which is computationally efficient and flexible to control over three types of network context through tunable parameters.

Our contributions can be summarized as:

- 1) **Novel Intent-aware Modeling.** We propose a novel intent-aware method to address the problem of diversified search on social networks, where nodes are represented as vectors, and the diversity of search result accounts for the overall query relevance and novelty among the retrieved nodes according to their representation vectors.
- 2) **Cooperative Network Representation.** We propose from the perspective of intent-aware, the *aspect2vec*, a novel network representation learning model, which represents nodes as vectors while cooperatively preserving network topological structure and node attributes.
- 3) **Scalable and Unified Random Walk.** We propose an attribute augmented random walks sampling algorithm to improve the effectiveness of the optimization, which samples edges, attributes and query-oriented proximity simultaneously via unified random walks.
- 4) **Evaluation.** We conduct extensive experiments to show the effectiveness of our method for diversified search tasks on real-world datasets, and find that our intent-aware solution significantly outperforms state-of-the-art diversification algorithms.

II. RELATED WORK

Search result diversification for documents. Search result diversification is widely being studied in the field of information retrieval. In order to tackle query ambiguity and result redundancy, diversification approaches in the literature often seek to achieve two goals: a high coverage of information needs and a high novelty among the retrieved documents, which can also be conflicting objectives [14]. Representative novelty-based algorithms include MMR [21], MaxSum [22] and MVA [23]. The majority of these methods adopt an implicit aspect representation, typically comprising the space of unique terms in a document corpus. Instead of implicitly modelling aspects, most of coverage-based algorithms directly maximize the coverage of multiple ‘facets’, which are commonly interpreted as aspects of the query according to a taxonomy [24], click logs and anchor text [25] or topic model [5]. Some hybrid models based on explicit query aspect representations have also been successfully investigated in the literature, including PM-2 [26] and xQuAD [27]. We also tackle this problem by explicitly modelling query aspects, but we focus on social network data, and our method

learns query aspects by the network representation learning model.

Search result diversification for networks. Enhancing diversity in searching for nodes in networks has been identified as an important retrieval and mining task. For example, DivRank [12] employs a time-variant random walks process to facilitate the rich-gets-richer mechanism in node ranking. DRAGON [13] optimizes a predefined direction-aware diversified goodness measure. Recently, a neighbour coverage based method has been proposed, which optimizes a bi-criteria objective of relevance and neighbour coverage [7]. The expanded relevance [28] improves this model by combining both relevance and diversity into a single function. However, all of these works optimize a global diversified goodness based on the premise of implicit query intent. In contrast, we tackle the problem in a different way that we infer query aspects and represent each node underlying the aspects of the query.

Network representation learning. NRL is a data embedding method to learn low-dimensional node representation, which preserves the structure of networks in their original space. To capture neighbourhood similarity and community membership, DeepWalk [18] learns network representations combining random walks and skip-gram model. LINE [17] extends to the model that has multiple context spaces for preserving both first-order and second-order proximity information between nodes. GraRep [15] learns the network representations using k -step loss functions that integrates local structural information and global structural properties of the network. Node2vec [16] uses flexible random walks over BFS and DFS to learn representations. SDNE [29] learns node representations for networks using a semi-supervised deep model with multiple layers of non-linear functions to capture the highly non-linear network structure. Some other NRL models focus on learning network representations preserving other context information of networks, such as labels [30], attributes [31] and communities [32]. Differently, our model learns representations of nodes under the specified query node to conduct diversified search task on social networks.

III. PRELIMINARIES

In this section, we first introduce our notations and terminologies, and formally define our problem to be addressed. Then, we briefly review two state-of-the-art diversification algorithms which are used in the second step of our method.

A. NOTATION AND TERMINOLOGY

In the remainder, we distinguish between *query*, *intent*, and *aspect*. A *query* is an expression of the information needs, which is represented as a node in this paper. The *intent* is the true information needs behind users who release a query. In some cases, the *intent* of a query user may be ambiguous, so they search from the information system and try to find out what information is they actually need. An *aspect* (sometimes called subtopic at the TREC Web track [5]) is an interpreta-

tion of the information need by the information system. How to the correctly interpret intents behind a query is the key problem of information retrieval systems.

We use $G = (V, E, W, A)$ to denote a social network with $|V|$ nodes and $|E|$ edges, where V is the node set, E is the edge set and A is a matrix recording node-attribute associations. Each edge $(u, v) \in E$ associates with a real positive weight $w_{uv} \in W$. The element $A_{ua} \in A$ denotes the binary-valued attribute weight indicating that attribute a associates with node u iff $A_{ua} = 1$. We let $A(u)$ denote the attribute set associated with node u and $|A|$ denote the total number of node-attribute associations.

B. PROBLEM FORMULATION

Based on the terminologies described above, we hold that a search result being aware of multiple intents of query is more hopeful to satisfy the information needs of users who have ambiguous intents. Building effective query aspects is of paramount importance to represent diverse query intents. In particular, we want such representation to be robust for any given query with different resolution and different number of aspects. Further, aspects should be meaningful to node attributes such as interest, institution and age, and to social relationships such as family, company and school. In general, there are two resources that can be used in our problem: edges and attributes. Thus, we define that an aspect of a query node refers to a feature subspace of a group of nodes, which are closely related to the query, tightly connected and share common attributes. In Fig. 2(a), we use a toy example to illustrate the multiple aspects of a query (NodeId:0). There are three distinct aspects collecting nodes with common features: nodes in group (1) are densely connected; nodes in group (2) share 5 common attributes with node 0; group (3) shares 2 attributes and densely connected.

We formally define the problem of **Intent-aware Diversified Search on Social Networks** as follows:

Given a social network G , a query node q , result size k , the problem is to search for a result set with k nodes that has: (1) the maximum relevance to q , (2) the maximum coverage w.r.t. the different aspects underlying q , and (3) the maximum novelty of the result set.

C. MAXSUM AND PM-2

There are basically two ways to make the search result diverse: a high novelty among the retrieved result and a high coverage of possible query aspects [14].

MaxSum is a novelty-based diversification algorithm that has theoretical guarantees in approximation [22], [33]. Given a query node q , and a fixed number k of candidate nodes, the goal of MaxSum is to find a subset $S \in V$ that satisfies:

$$\begin{aligned} \max_{S \subset V} \quad & f(S) = \sum_{u \in S} \text{sim}(u, q) - \lambda \sum_{u, v \in S} \text{sim}(u, v) \\ \text{s.t.} \quad & |S| = k, \end{aligned} \quad (1)$$

where $\text{sim}(\cdot, \cdot)$ is the similarity function, and $\lambda > 0$ is a free parameter specifying the trade-off between relevance

and dissimilarity. In this paper, we directly apply the negative squared Euclidean distance as the similarity function. By using this algorithm, the diversification problem can be tackled by attaining a high relevance of the possible aspects underlying q while ensuring the novelty (i.e. eliminating redundancy). Although maximizing the objective $f(S)$ is NP-hard, there is a simple linear time greedy algorithm that iteratively selects node with the maximum marginal gain of $f(S)$, which achieves a 2-approximation solution for the problem [33].

PM-2 is a proportionality-based diversification model derived from the Sainte-Laguë method aiming at finding a proportional coverage of each aspect of the query [26]. It first computes a proportionality quotient $\text{quot}(s|q)$ for each query aspect s , according to: $qt(s|q) = \frac{v_{i|q}}{2s_{i|q} + 1}$, where the numerator represents the number of "votes" received by the i -th aspect and $s_{i|q}$ represents the percentage of "seats" the i -th aspect received. Then the "seat" will be awarded to the aspect with the largest quotient in order to best maintain the proportionality of the list, and filled by the one that is not only relevant to current aspect but also to other aspects:

$$\begin{aligned} v^* \leftarrow \arg \max_{v_j \in V} \lambda qt[s^*|q]P(v|s^*, q) \\ + (1 - \lambda) \sum_{i \neq i^*} qt[s|q]P(v|s, q), \end{aligned} \quad (2)$$

where $P(v|s, q)$ measures the coverage of node v with respect to s for a given q . In our experiments, $P(v|s, q)$ is measured by the element-wise relevance of each aspect rather than the probability relevance, since the representation of nodes is real-valued in our intent-aware method. Parameter λ in this equation trades the relevance to the awarded aspect and other aspects. Accordingly, the search result can be improved in diversity by selecting nodes proportionally covering the query aspects.

IV. OUR METHOD

A. OVERVIEW

To solve the problem, we propose an *intent-aware* method that first represents possible query aspects and nodes as vectors and then selects representative nodes according to these embedding vectors. Our method consists of two main steps and is described in Algorithm 1.

Algorithm 1 Overview of our intent-aware method.

Input: G, k, q .

Output: Top- k diversified nodes.

- 1: Learn representations of query aspects and nodes F ;
 - 2: Perform diversification based on F ;
-

In the first step, our algorithm uses a new NRL model to learn a representation matrix $F \rightarrow \mathbb{R}^{|V| \times d}$ that describes the feature representation of each node, where d is the size of latent dimensions. It means that each aspect is expressed by a subset of dimensions and each dimension contributes to a combined feature subspace. In particular, our NRL approach

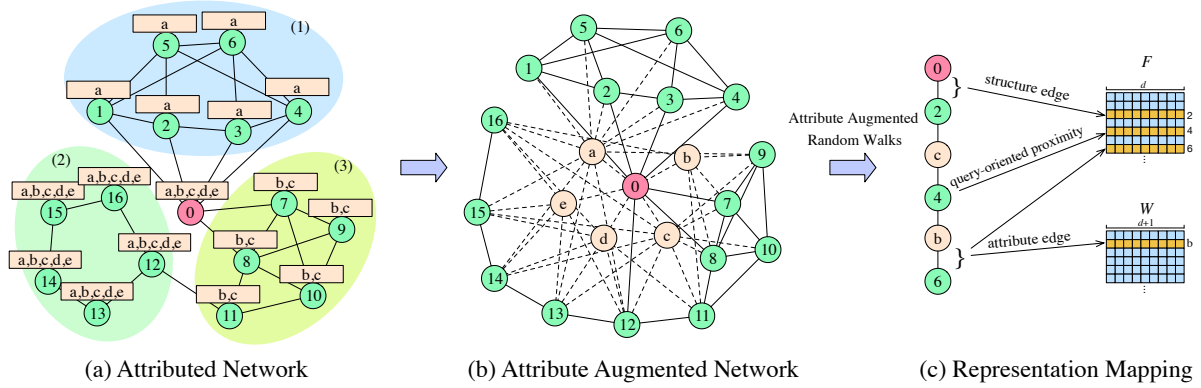


FIGURE 2: Overview of aspect2vec: a toy network with 17 nodes and 5 attributes.

seeks to learn representations for networks that not only satisfy characteristics of adaptability, community aware, low-dimension and continuity [18], but also the followings:

1) **Intent-aware**: All possible latent aspects of the query should be well expressed in the vectors among the whole network. The embedding vectors of other nodes should be mapped w.r.t. these aspects so that the aspect relevance of the retrieved nodes to the query could be well measured.

2) **Heterogeneity**: Aspects of nodes are heterogeneously formed by structure feature, attribute feature and their combination. Our NRL model should be capable of jointly leveraging the heterogeneous and noisy information to compensate each other towards an effective and robust embedding representation.

We will detail the design and optimization of our new NRL model in the next two subsections.

In the second step, the diversification subroutine uses the output of first step as input to conduct a diversity-oriented ranking and returns the result. In this paper, we consider **MaxSum** [22] and **PM-2** [26], two state-of-the-art search result diversification algorithms, as the diversification part of our method to evaluate the performance of our NRL model.

B. ASPECT2VEC

To better tackle the diversification problem, a desirable NRL model for real-world social networks should satisfy the following requirements: first, it must be able to preserve the structure relationships of nodes; second, it must be able to preserve the attributes of nodes; third, it must be able to represent aspects of the query as much as possible.

We develop *aspect2vec*, a scalable and flexible NRL model that satisfies the above requirements. The model aims at obtaining embedding representations of all the nodes in G , denoted as F with the element F_u being the embedding representation for node u . We achieve this by jointly modelling three components of contextual information: network structure, node attributes and the query-oriented proximity. Modelling network structure and attribute ensures that the learned embedding space can well reflect the community structures in the original network, while modelling query-

oriented proximity assures that these communities are locally mapped underlying the query (i.e. query aspects). In what follows, we describe the components of aspect2vec in details.

Modelling network structure. To preserve the network structure, edges of pairwise vertices must be preserved, since two users in real-world social networks are similar if they are linked by a relational edge. We assume that two nodes u and v represented by F_u and F_v are connected with the following probability:

$$P_{uv} = \sigma(F_u^T F_v), \quad (3)$$

where $\sigma(\cdot)$ is a logistic function defined as $\sigma(x) = (1 + e^{-x})^{-1}$. To preserve edges of the network, we can naturally minimize the Kullback-Leibler (KL) divergence between P_{uv} and the empirical distribution $\hat{P}_{uv}$:

$$\mathcal{L}_E(F) = D_{KL}(\hat{P} \parallel P). \quad (4)$$

Here the empirical distribution $\hat{P}_{uv}$ is defined as the normalized edge weight w_{uv} . The objective function in Eq. 4 can be simplified as minimizing the following function:

$$\mathcal{L}_E(F) = - \sum_{(u,v) \in E} w_{uv} \log P_{uv}. \quad (5)$$

Modelling node attributes. The idea of preserving edges could also be extended to model node attributes. Similar to modelling network structure, we perform the same procedure on node attribute affinity by defining an attribute latent weight vector. Specifically, given a node u with an attribute a , we use the logistic model with u 's embedding vector F_u as input features to predict the probability of u associating a :

$$Q_{ua} = \sigma(W_a^T F_u + b), \quad (6)$$

where W_a is the logistic weight factor and b is the bias. Thus, the loss function is defined as follows:

$$\mathcal{L}_A(F) = D_{KL}(\hat{Q} \parallel Q), \quad (7)$$

where $\hat{Q}_u$ is the empirical distribution of attribute weight of u , which can be simply set as $\hat{Q}_{ua} = A_{ua}$. It is worth

mentioning that $\hat{Q}_{ua}$ is applicable for both binary-valued and real-valued attributes, but in this paper we mainly focus on binary-valued attributes. In addition, to prevent the weight factor W_a from being too large and avoid overfitting, we introduce an l_2 -regularization term $\|W_a\|_2$. Thus, the objective function of Eq. 7 can be calculated with parameter ρ as:

$$\mathcal{L}_A(F) = - \sum_{u \in V} \sum_{A_{ua} \in A(u)} A_{ua} \log Q_{ua} + \rho \|W\|_F^2. \quad (8)$$

Modelling query-oriented proximity. Nodes' embedding vectors constructed by the above two models are based on global structural and attributed information. In our approach, we wish to learn node representation underlying the query aspects, so that important aspects of the query node can be well expressed and other nodes can be mapped into the feature subspaces of the query node. To achieve this goal, we construct an objective function to learn node representations while preserving the similarities of nodes to the query, i.e., *the query-oriented proximity*. Then the probability of predicting the similarity of a node u to query q , given F_u and F_q , can be defined as a softmax function:

$$R_{qu} = \frac{\exp(F_u^T F_q)}{\sum_{v \in V \setminus q} \exp(F_v^T F_q)}. \quad (9)$$

Similarly, the loss function for this goal can be defined as:

$$\mathcal{L}_S(F) = D_{KL}(\hat{R} \| R) = - \sum_{v \in V \setminus q} \hat{R}_{qv} \log R_{qv}, \quad (10)$$

where $\hat{R}_{qv}$ is the empirical distribution of similarity between query q and node v . The loss function indicates that the more similar they are, the closer their embedding vectors will be. In other words, representation vectors of relevant nodes will be locally mapped to the aspect spaces of the query according to their query-oriented proximity.

Thus, combining Eq. 4, Eq. 7 and Eq. 10, our optimization problem that we aim to solve is:

$$\min \mathcal{L}(F) = (1 - \alpha) \mathcal{L}_E(F) + \alpha \mathcal{L}_A(F) + \beta \mathcal{L}_S(F), \quad (11)$$

where α and β are the free parameters.

C. MODEL OPTIMIZATION

To obtain optimal F for $\mathcal{L}(F)$ in Eq. 11, we use the Asynchronous Stochastic Gradient algorithm (ASGD) [34] due to the sparsity of the optimization problem, which means that most gradient updates only modify a small portion of the variables. For objective $\mathcal{L}_E(F)$ and $\mathcal{L}_A(F)$, we can directly optimize them by iteratively updating the embedding vector F according to the distributions of edges and attributes, which needs at least $O(|E| + |A|)$ time.

However, it is usually intractable to directly optimize R_{qu} in Eq. 10, because the denominator of Eq. 9, $\sum_{v \in V \setminus q} \exp(F_v^T F_q)$, is expensive to compute for large networks. To alleviate this bottleneck, we approximate the

probability distribution using the Negative sampling (NEG) approach [35], which has been shown to be effective in data embedding models [16], [17], [35]. NEG approximately maximizes the log probability of the softmax into a probabilistic classification problem to distinguish samples from the empirical distribution and a noise distribution. More specifically, it specifies the following objective function for each node:

$$\log \sigma(F_u^T F_q) + \sum_{i=1}^K \mathbb{E}_{v_i \sim p_{|V|-1}(v)} [\log \sigma(-F_{v_i}^T F_q)], \quad (12)$$

where K is the number of negative samples and $|V|$ is the number of nodes. To train the model, we draw samples from the original network and treat them as positive and negative samples, according to their noise distributions of the three types of context information.

Training model via random walks. Similar to the language Modelling convention [35], we now need a "corpus" as input of aspect2vec to train our model. Given a social network, aspect2vec considers three types of contextual information as its corpus: structure, attribute and query-oriented proximity. For structure context, a native method is to sample positive samples from the original edges with the sampling probabilities proportional to the original edge weights. For attribute context, we can adopt a similar strategy that views each attribute association as an edge and samples such edges according to their attribute weights. For query-oriented proximity context, a simple treatment is to obtain a query-oriented proximity distribution from the pre-computed similarity matrix by some similarity measures, such as Personalized PageRank (PPR) [36], SimRank [37] or Panther [1]. Thus, a straightforward solution is to optimize the objective of Eq. 11 by deploying three separate sampling procedures. However, this separate sampling method not only makes model complicated, but also significantly increases the total cost, as similarity estimation itself is a high computational problem [38], especially in very large social networks. According to Algorithm 1, to obtain diversified results for an arbitrary given query, we need to re-encode the entire network w.r.t. the query, which is costly for frequent searches in large scale networks.

To address this issue, we design a flexible random walks strategy called *attribute augmented random walks* based on the *attribute augmented network*, that generates context of edges, attributes and query-oriented proximity efficiently through single-source random path sampling.

Attribute augmented network. The attribute augmented network is commonly used for attributed network to combine edges and attributes into one unified feature [39]. Formally, an attribute augmented network is denoted by $G_A = (V \cup V_A, E \cup E_A, W \cup W_A)$ where V_A is a ground set of attributes in G . $a \in V_A$ is called attribute node in G_A . $E \subseteq V \times V$ is the structure edge set and $E_A \subseteq V \times V_A$ is the attribute edge

set. An edge $(u, a) \in E_A$ iff $A_{ua} = 1$ in G . W_A measures the binary weight of attribute edges.

Transition probability. The transition matrix of G_A , i.e., P_{G_A} , is a $|V \cup V_A| \times |V \cup V_A|$ matrix that defines the transition probability of each random walk.

From a structure node u , it can either walk to a structure node or an attribute node. Unlike the prior work [39], [40], the probability of choosing the node type of the next target is defined as the value of the tradeoff factor α in Eq. 11, i.e., structure edges take $1 - \alpha$ transition probability and attribute edges take α transition probability.

We normalize ω_{uv} and ω_{ua} by $\frac{\sum_{v \in N(u)} \omega_{uv}}{\sum_{v \in N(u)} \omega_{uv} + \sum_{a \in A(u)} \omega_{ua}} = 1 - \alpha$, where $\omega_{uv} \in W$ is the structure edge weight, and $\omega_{ua} \in W_A$ is the attribute edge weight. $N(u)$ and $A(u)$ are the adjacent node set and the attribute set of node u respectively. Then the transition probability from structure node u to structure node v is:

$$p(u, v) = \begin{cases} \frac{\omega_{uv}}{\sum_{w \in N(u)} \omega_{uw}}, & \text{if } (u, v) \in E \\ 0, & \text{otherwise} \end{cases} \quad (13)$$

The transition probability from a structure node u to an attribute node a is defined as:

$$p(u, a) = \begin{cases} \frac{\omega_{ua}}{\sum_{w \in A(u)} \omega_{uw}}, & \text{if } (u, a) \in E_A \\ 0, & \text{otherwise} \end{cases} \quad (14)$$

The transition probability from an attribute node a to a structure node u through an attribute edge is:

$$p(a, u) = \begin{cases} \frac{1}{|V_a|}, & \text{if } (a, u) \in E_A \\ 0, & \text{otherwise} \end{cases} \quad (15)$$

where $|V_a|$ is number of the nodes that have attribute a . The transition probability between any two attribute nodes is 0, since there is no edge between two attribute nodes.

Attribute augmented random walks. To sample a path, the random walks generator selects the query node q as the starting point and then conducts random walks for l steps according to their transition probabilities.

Different from DeepWalk that samples each path starting with a random node, we start with the query node, because this is an effective way to estimate the query-oriented similarity of nodes according to their co-occurrence appearing frequency among all the paths [1]. Nodes with similar distributions over the contexts are assumed to be similar [17], which means high query-oriented similarity nodes will be more easily represented on the feature spaces of the query. In addition, structure edges and attribute edges among the paths also reflect the local context information of corresponding nodes. Therefore, our random walks strategy can capture the three types of contextual information mentioned above.

There are several benefits of our random walks strategy over the separate sampling procedure.

- 1) It provides a flexible mechanism to sample edges, attributes and query-oriented proximity simultaneously by a unified random walk.
- 2) It merges the tradeoff parameter α of loss function Eq. 11 into the random walks process. The bigger the value of α is, the more likely the random walks process will pick an attribute edge. With this path-sampling treatment, the overall objective function Eq. 11 remains the same.
- 3) Our random walks strategy is a local exploration from the query node. The more relevance exists between the nodes and the query, the more frequency nodes will appear on the path. In other words, it could reduce the numbers of update of irrelevant nodes, which have less influence to diversified search result.
- 4) Our random walks strategy is easy to parallelize. It is also possible to accommodate small changes in the network without needing to perform global recomputation. We can iteratively update the learned vectors with new random walks from the changed region in time sublinear to the entire network.

Training aspect2vec. For training the model, we sequentially pick an edge from sampled random walks and then use it to update the representation vectors. The pseudocode for training aspect2vec is provided in Algorithm 2. Specifically, Algorithm 2 iterates over all edges in collocations of random walks $\mathcal{W}$ (lines 5-15). For each edge in a walk, we obtain a gradient $e0$ based on $\mathcal{L}_E(F)$ or $\mathcal{L}_A(F)$ depending on α , the probability of choosing a structure edge or an attribute edge (lines 7-11). We also obtain $e1$, the gradient based on $\mathcal{L}_S(F)$ (lines 12-13). Then representation vectors will be updated by using ASGD [34] algorithm with the combined gradient ($e0 + \beta e1$). We use the alias table method [41] to draw a sample according to the node frequency among all the random paths, which takes only $O(1)$ time when repeatedly drawing samples from the same discrete distribution.

Fig. 2(b) shows an example describing how to build the corresponding attribute augmented network from the toy network in Fig. 2(a). In Fig. 2(c), we show a sampled random path $\{0, 2, c, 4, b, 6\}$ with length $l = 5$, consisting of five edges $\{(0, 2), (2, c), (c, 4), (4, b), (b, 6)\}$ and three structure nodes $\{2, 4, 6\}$. These edges and the frequency of nodes are then used to obtain gradient $e0$ and $e1$ respectively to update the corresponding embedding vectors.

The overall time complexity of aspect2vec is $O(dKlr)$, which is linear to the number of random walks r and walk length l , and does not depend on the number of nodes $|V|$.

V. EXPERIMENTS AND RESULTS

Here, we detail experimental setup and report the results.

A. EXPERIMENTAL SETUP

Research Questions. The research questions guiding the remainder of the paper are:

(RQ1) Do node representations generated by aspect2vec

Algorithm 2 Training Algorithm for aspect2vec**Input:** G , α , β , q , and number of random walks $-r$.**Output:** Node embedding representations F .

```

1: Construct the attribute augmented network  $G_A$ ;
2: Initialize the transition matrix  $P_{G_A}$ ;
3:  $\mathcal{W} = \text{RandomWalk}(G_A, P_{G_A}, q, \alpha, r)$ ;
4: Compute Alias table  $\mathcal{S}$  from node frequency of  $\mathcal{W}$ , and Initialize  $F$ ;
5: for each  $\mathcal{W}_i \in \mathcal{W}$  do
6:   for each  $(u, v) \in \mathcal{W}_i$  do
7:     if  $(u, v)$  is a structure edge then
8:       Draw  $K$  negative nodes randomly, and Compute the gradient  $e0(v)$ ;
9:     else
10:      Draw  $K$  negative attributes randomly, and Compute the gradient  $e0(v)$ ;
11:       $\{(u, v) \text{ is an attribute edge, w.l.o.g. assume } v \text{ is a structure node.}\}$ 
12:    end if
13:    Draw  $K$  negative nodes according to  $\mathcal{S}$ , and Compute the gradient  $e1(v)$ ;
14:     $F_v = F_v - \eta(e0(v) + \beta e1(v))$ ;
15:   end for
16: end for
17: return  $F$ ;

```

meet the requirements of the diversification task?

(RQ2) Does our intent-aware method outperform the state-of-the-art diversification methods?

(RQ3) Do attributes help improve the diversification performance?

(RQ4) What is the impact of free parameters α , β and l in our algorithms?

(RQ5) What is the time efficiency of our algorithms?

Datasets. We consider three real-world social networks with their brief statistical information listed in Tab. 1.

- 1) The Facebook dataset is downloaded from SNAP². This dataset is built from profile and relation data of 10 users in Facebook, and the attributes are constructed by their profiles. There are 193 social circles in this network, which were manually identified by the 10 users via choosing which community their friends belonged to, such as universities, sports teams, relatives, etc. [42]. We consider these social circles as the ground truth aspects of the 10 users.
- 2) The DBLP dataset is crawled from the DBLP public bibliography data³. We build a coauthor network extracted from the paper data of top 172 computer science conferences (Tiers A and B from China Computer Federation⁴). We treat each author as a node, collaboration of paper as an edge. The attributes in DBLP dataset are constructed by the extracted keyterms of their papers. The conferences in which authors have published papers are viewed as the ground-truth aspects.

²<http://snap.stanford.edu/data/>

³<http://dblp.uni-trier.de/xml/>

⁴<http://www.ccf.org.cn/>

TABLE 1: Statistics of real-world datasets

Datasets	Nodes	Edges	Attributes	Aspects
Facebook	4,039	88,234	1,406	193
DBLP	73,242	373,797	122,043	172
Flickr	80,513	5,899,882	-	195

TABLE 2: Comparison methods

Intent-aware methods			Implicit intent methods	
	MaxSum	PM-2		
DeepWalk	DWMS	DWPM2	Exp2	RelExp2
LINE	LNMS	LNPM2		
aspect2vec	A2VMS A2VMSNA	A2VPM2 A2VPM2NA		

- 3) The Flickr dataset is collected by crawling from the Flickr website⁵. Edges of this network are built from users' friendships, and their group memberships are viewed as the ground-truth aspects.

The three networks are undirected and their edge weights are set to 1. The Flickr network is a non-attributed network.

Baselines. To evaluate the performance of our intent-aware method, we consider several baselines, including three state-of-the-art NRL methods (i.e. DeepWalk [18], LINE [17] and node2vec [16]) and two implicit intent diversification algorithms (i.e. Exp2 [7] and RelExp2 [28]). We use A2VMS, A2VPM2, A2VMSNA and A2VPM2NA to denote the attributed and the non-attributed versions of our intent-aware method based on MaxSum [33] and PM-2 [26] respectively. As our intent-aware method builds on both NRL and diversification models, we also consider four intent-aware baselines (i.e. DWMS, DWPM2, LNMS and LNPM2), which integrate DeepWalk and LINE into MaxSum and PM-2. Tab. 2 summarizes their combinations and abbreviations according to a confusion matrix.

Evaluation Metrics. We evaluate our intent-aware method against several diversification metrics, which are widely used in tasks of search result diversification on networks [7], [28] and on documents [43].

- 1) *Density* (Den) [12]. The density metric is equal to the number of edges existing in the graph divided by the maximum possible number of edges of the graph. A smaller density means less structural relevance among the nodes, which implies higher novelty of the result.
- 2) *Group coverage* (GC). Group coverage quantifies the number of aspects covered by the result nodes [7]. The more aspects covered by the result nodes, the better diversity of the result that would meet the information needs for the over-all users.
- 3) *S-recall* (SR) [43]. S-recall quantifies the amount of groups covered by the top k nodes that are relevant to the query, calculated by: $SR = \frac{\bigcup_{v \in S_q} (A_q \cap A_v)}{A_q}$, where

⁵<http://socialcomputing.asu.edu/>

S_q is the top- k search result set and $\mathcal{A}_v$ is the number of aspects covered by node v . This metric measures whether the covered aspects is actually useful or not for the query.

We note that some intent-aware measures in textual diversification researches, such as α -nDCG [44], MAP-IA [24] and ERR-IA [45], are not included to the discussions, but they are important measures for evaluating the diversity of the results when the data and queries have some already known relevance assessments. However, unlike the textual search that has public test collections, the evaluation benchmarks for search result diversification on networks are not available as far as we know.

Parameter Settings. The parameter settings of η , d , and K used for aspect2vec are in line with typical values used for LINE [17] and node2vec [16]. Specifically, the learning rate η for ASGD is initially set to 2.5% at the beginning of the training and then decreased linearly with the number of random paths that have been used. For fair comparisons, the dimension d of all the embedding algorithms is set as 128. Other default settings include: the number of negative samples $K = 5$, the walks length $l = 80$ for node2vec and DeepWalk, and $l = 20$ for aspect2vec.

B. RESULTS

In the following, we discuss and analyze our experimental results and answer the research questions **RQ1** to **RQ5**.

Visualization on Facebook data. To answer research question **RQ1**, we focus on visualizing the learned vectors of the comparison NLR models for Facebook dataset. We take the node #1912 as a query node, and feed learned representations by aspect2vec and other compared methods: DeepWalk, LINE and node2vec, into the standard t-SNE tool [46] to lay out the representation vectors of different kinds of nodes on 2-D space. Here we set the parameters of node2vec, (p, q) , to be $(1, 0.5)$, as in this setting node2vec exhibits community characteristic [16]. According to Fig. 3(a-c), the layouts of nodes by DeepWalk, LINE and node2vec show distinct blocks and most social circles nodes (colored) of the query node are clustered together, which means that these models can preserve the community structure of the network. However, these colored blocks are distributed in all parts of the plane and seem to have no strong ties to the query. It means that these models fail to capture the local aspects of the query. In contrast, we can see from Fig. 3 (d) that in aspect2vec ($\alpha = 0.7$, $\beta = 0.6$), the social circles are scattered very closed to the query node and show clear boundaries between different colors, indicating that the local aspects of the query are expressed by our aspect2vec. From this subfigure, we can also see that the non-social circle nodes still do not exhibit in a block form. The reason is that, compared to #1912's social circle nodes, these non-social circle nodes do not frequently appear in the sampled paths yielding less vector update. Although other compared methods (Fig. 3 (a-c)) also encode

communities of the network, their query node are merged into one of the communities due to the resolution effect and some social circles of the query are laid far away from the query node. It is known that a desirable diversified search result for a query node in Facebook network should be related to the query and cover as many social circles as possible. From the visualizations of the three compared methods, it is hard to distinguish such diversified nodes according to these representations by an existing diversification algorithm.

Diversification performance. Then, we turn to research questions **RQ2** and **RQ3**. In this set of experiments, we randomly generate 100 queries, obtain top 10 and top 50 result nodes, and report the average performance on the three diversification metrics in Tab. 3. We make the following observations.

First, A2VPM2NA and A2VPM2 achieve the best performance in SR metric on pure network Flickr and on attributed network DBLP respectively. This demonstrates that our intent-aware method integrating PM-2 is fairly effective for covering query aspects in the diversified search task.

Second, we can see that in DBLP network our A2VMS and A2VPM2 outperforms A2VMSNA and A2VPM2NA respectively in all the metrics, which demonstrates that considering node attributes do help improve the diversification performance.

Third, the ten intent-aware methods show lower density than the two implicit intent methods (Exp2 and RelExp2) in most cases, and the MaxSum-based algorithms all outperform the PM-2-based algorithms on Den metric. Recall that the lower density implies that the result nodes are more novel, which indicates that our intent-aware methods, especially the MaxSum-based algorithms, are more capable to eliminate redundancy in diversified search task.

Fourth, we can also find that in Flickr the performance of GC reports the opposite result comparing with the density performance: Exp2 and RelExp2 outperform the others in most cases. More specifically, in Flickr RelExp2 achieves the best performance on both GC@10 and GC@50, and in DBLP Exp2 achieves the best and second best on GC@10 and GC@50 respectively. That is not a surprising result, because Exp2 and RelExp2 try to find the result by maximizing the coverage of their global diversity metric. However, they show relative underperformance in SR metric, which suggests that results generated by Exp2 and RelExp2 may, in large part, be of no interest to the query.

In summary, all these results demonstrate the effectiveness of our aspect2vec model and our intent-aware method in diversified search task.

Parameter sensitivity. RQ4: In this subsection, we investigate the effects of parameters α , β and l in our algorithms.

As discussed in our model, α balances the contributions of network structure and node attributes. We examine how the different choices of α affect the formation of different aspects in our toy network (Fig. 2(a)). Fig. 4 visualizes the

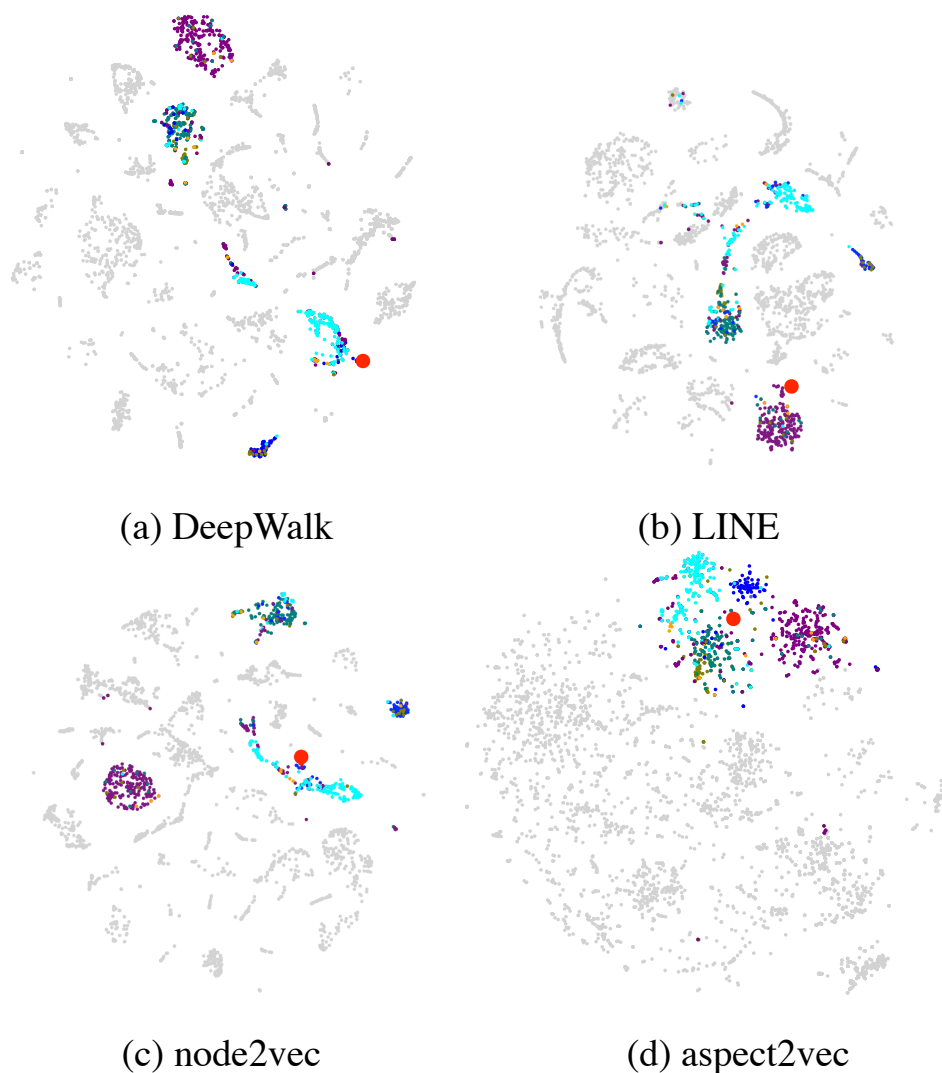


FIGURE 3: Visualization of representation vectors on Facebook network. The big red node represents the query node (the user with ID #1912); other colored nodes represent its social circle friends; the gray nodes represent those not within the user #1912's social circle.

representation vectors in 2-D space by different α . We can see that nodes in group (2), the aspect that is more relevant to attributes, are relatively scattered by $\alpha = 0.2$, and they become denser as α increases. In contrast, nodes in group (1), the aspect that is more relevant to edges, are relatively denser than other two groups, and they gradually spread apart as α increases. The densities of the three groups are relatively indistinguishable when $\alpha = 0.6$, $\beta = 0.7$. It means that parameter α flexibly controls the weight between structure aspects and attribute aspects.

Then we show how the value of β affects the performance in Fig. 5(a). The parameter of β balances the weight of the overall network context (edges and attributes) and the query-oriented proximity context. We can see from Fig. 5 (a) that A2VMS and A2VPM2 show a same trend in the SR metric. When $\beta = 0.1$, their performance of S-recall is not good,

because in this case some important aspects of the query cannot be well represented. Then the S-recall scores get higher as β increase and reaches the best performance at $\beta = 0.7$. However, when β is too large, the performance decreases too, since in this case our model gives less weight to capture the structure and attribute characters. This result suggests that the query-oriented proximity is the key to capture the local aspects of the query.

Fig. 5(b) shows the S-recall performance of A2VMS and A2VPM2 on DBLP network by varying path length l from 10 to 80. We see that too small l ($l \leq 5$) would result in inferior performance. When increasing l up to 20, their performance almost become stable.

Scalability Finally, we turn to answer research question **RQ5**. To study the scalability, we obtain the node repre-

TABLE 3: Diversification performance. The best and the second best performing runs per metric per dataset are marked in boldface and underlined, respectively. Statistical significances of performance improvements between attributed and non-attributed version of our intent-aware methods are marked in the lower right hand corner.

Data	Algorithms Metrics	Exp2	RelExp2	DWMS	DWPM2	LNMS	LNPM2	A2VMSNA	A2VMS	A2VPM2NA	A2VPM2
Flickr	Den@10	0.552	0.686	0.220	0.252	0.220	0.244	0.112	N.A.	<u>0.188</u>	N.A.
	Den@50	0.284	0.346	0.224	0.320	0.188	0.235	<u>0.208</u>	N.A.	<u>0.244</u>	N.A.
	GC@10	0.157	0.186	0.102	0.138	0.122	0.152	0.144	N.A.	<u>0.172</u>	N.A.
	GC@50	0.358	0.372	0.223	0.262	0.238	0.264	0.295	N.A.	<u>0.358</u>	N.A.
	RC@10	0.229	0.182	0.226	0.282	0.192	0.220	<u>0.308</u>	N.A.	0.392	N.A.
	RC@50	0.503	0.628	0.556	0.595	0.520	0.535	<u>0.682</u>	N.A.	0.750	N.A.
DBLP	Den@10	0.667	0.333	<u>0.024</u>	0.025	0.182	0.220	0.058	0.021 ▲	0.667	0.305▲
	Den@50	0.344	0.096	<u>0.030</u>	0.048	0.038	0.058	<u>0.025</u>	0.010 ▲	0.112	0.096▲
	GC@10	0.323	0.222	0.162	0.200	0.152	0.222	0.182	0.234▲	0.265	<u>0.305</u> ▲
	GC@50	<u>0.610</u>	0.480	0.445	0.485	0.466	0.525	0.462	0.535▲	0.605	0.664 ▲
	RC@10	<u>0.482</u>	0.320	0.436	0.485	0.440	0.493	0.515	0.620▲	<u>0.682</u>	0.725 ▲
	RC@50	0.719	0.646	0.731	0.793	0.793	<u>0.850</u>	1	1	1	1

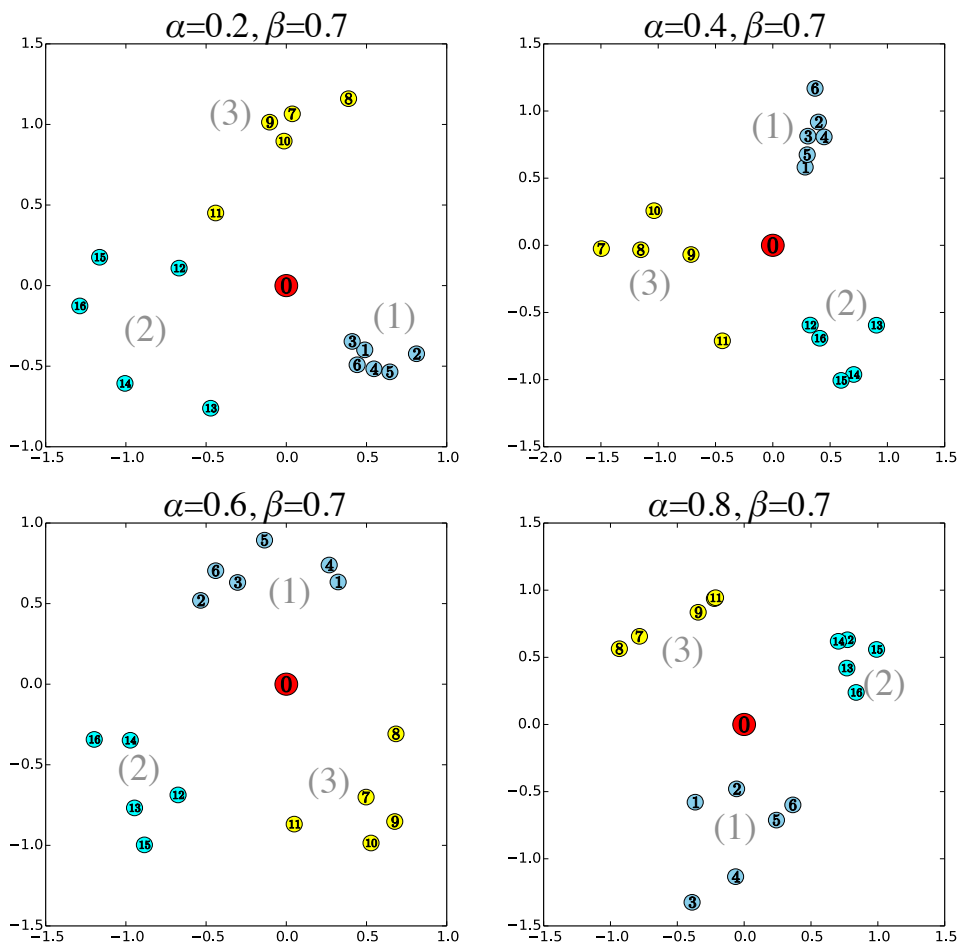


FIGURE 4: Impact of α . Representation of the toy network in Fig. 2(a). The three groups of nodes differ in colors.

sentrations of DBLP network by using aspect2vec with fixed parameter values over different random walks sizes and different dimensions. We ran the algorithms on a PC with 16GB of RAM and an Intel Core i7-3770 CPU 3.40GHz. Here we set $k = 50$, and similar results can be observed in other k .

Fig. 6(a) shows the running time of aspect2vec, A2VMS and A2VPM2 over $r * l$. According to Fig. 6(a), the time cost of the three algorithms is close to linear. Fig. 6(b) shows the running time over different dimensions. As can be seen in the subfigure, the running time increases approximately linearly

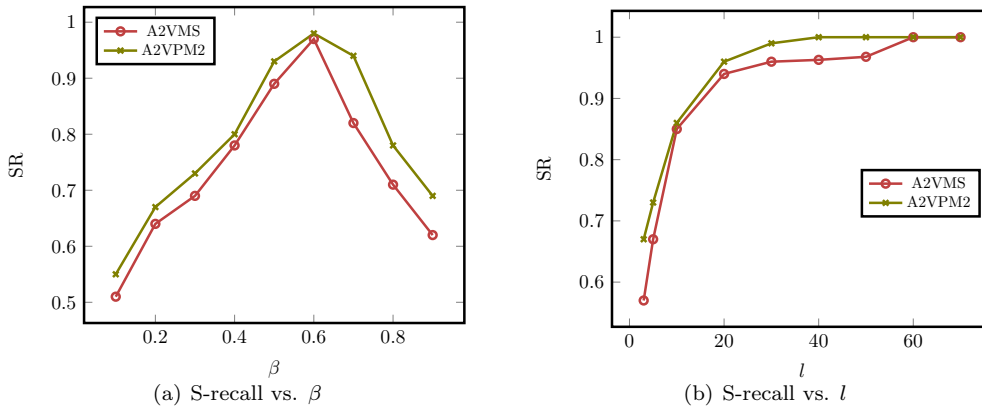


FIGURE 5: Impact of β and l . S-recall performance of A2VMS and A2VPM2 by different β and l on DBLP network

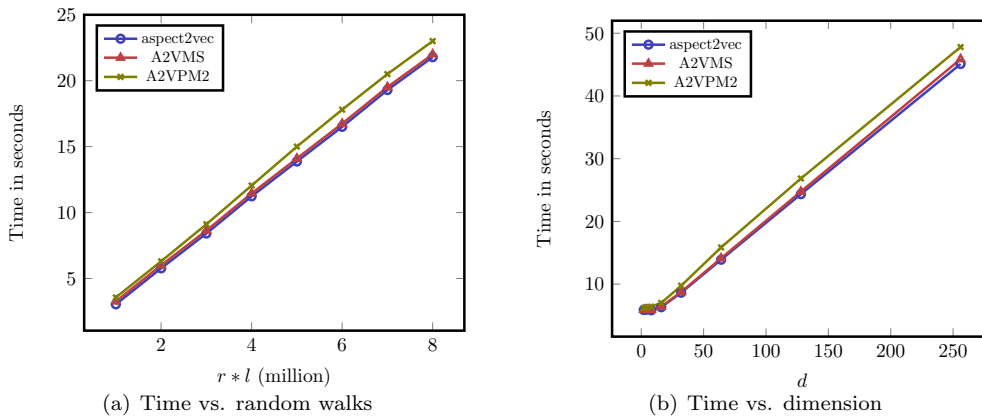


FIGURE 6: Running time of aspect2vec, A2VMS and A2VPM2 by different number of random walks and dimension on DBLP network.

as the dimension increases. All these results show that our aspect2vec model is scalable to large scale social networks.

VI. CONCLUSIONS

In this paper, we have studied the problem of search result diversification on social networks. To tackle the problem, we propose a novel intent-aware method that first explicitly represents query aspects and nodes as unified vectors by aspect2vec, a new NRL model, then selects representative nodes by jointly maximizing the overall relevance of query aspects and the diversity among the result nodes based on these vectors. Our aspect2vec model projects a social network into a unified low-dimensional embedding space w.r.t. the query aspects while jointly preserving the community structure and node attributes. To further accelerate the training of aspect2vec, we have proposed an attribute augmented random walks strategy for sampling samples among three contexts efficiently.

In our experiments, we have evaluated the performance of our intent-aware method and the baseline algorithms against

three diversification metrics on three real-world social networks and have demonstrated that our approach is very fast and significantly outperforms state-of-the-art baselines on search result diversification task. Visualization on Facebook data shows that our aspect2vec model properly maps relevant nodes into vectors close to the subspace of the query node, which is helpful for making the possible important query aspect explicitly expressed. We have also shown that the use of node attributes can enhance the performance of diversifying search result. The result nodes from intent-aware based methods that represents the possible query aspects and nodes as vectors while preserving structure and other side information, can have more informative explanations of users' query preferences than that from simple diversified goodness ones used in previous literature.

There are two main limitations in our proposed approach: a) it does not take the fact that edges and attributes are evolving over time in many real-world social networks into account; b) it only incorporates the binary attribute information in the network representation model, while some other

latent diversified features such as documents and diffusion influence are of great value to be integrated in the model. As to future work, we aim to incorporate documents and diffusion influence into aspect2vec model, which will be used to interpret semantic and influence of each aspect, and consider dynamic embedding modelling.

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